

Crowdsourcing Data for Safer Travel Experiences for Women in India

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ABSTRACT

In 2018, the Thomas Reuters perception survey ranked India as the most dangerous country for women due to the high risk of sexual, and non-sexual violence, and harassment [2]. In a study conducted by Jagori and UN Women in Delhi, it was observed that the highest number of incidents reported by women occurred on buses, other public transportation, and on the streets [3]. With the rise of women traveling and entering the workforce, it's critical to observe how Information and Communications for Technology and Development (ICTD) may be designed to increase female safety while traveling. Through a diary study, competitor analysis, and user interview methodology, our team uncovered unique challenges in public transportation, social and gender role expectations, and familial interventions that shape the safety of women in Southern India. Informed by our research, we created Navi.io, a mobile safety application that enables users to feel safer and make informed decisions when traveling by utilizing crowdsourced data and connecting with contacts in transit.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • HCI design and evaluation methods;

KEYWORDS

Women Safety, India, Navigation, Mobile Application, Location Sharing, Public Transportation

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1 INTRODUCTION

Women in India face significant safety concerns like a rise in the rates of kidnappings, abductions, rapes, domestic violence, dowry deaths, and assaults [5]. Most notably, studies have shown high rates of crime against women reported on public transportation which is exacerbated by India's large reliance on public transportation. The topic of women's safety has been a major concern in India

after incidents like the 2012 Nirbhaya case caused a public uproar across the country [7]. Since then, the government has implemented policies to address female safety, such as female segregated sections on public transit, longer jail sentences for assault, and helplines. However, these efforts toward improving women's safety have been limited and inconsistent, leaving women still feeling unsafe and unsupported in their communities.

In a 2013 study conducted by the UN Women's Safe Cities Initiative, 92% of women between the ages of 16-49 reported experiencing at least one form of sexual violence [1]. However, in the same study, 40-71 percent of women exhibited limited response strategies, when encountering acts of sexual violence. Many women simply did not know how to respond. Few reported incidents to the police or family members due to the pervasive culture of victim blaming. This finding corroborates another study that found that the major reasons why women do not report the incident include (1) incidents perceived as a minor offense, (2) fear of attracting attention, (3) fear of retaliation, (4) fear of hurting family reputation, and (5) fear of being blamed.

For these reasons, our team was motivated to study the intersection of public transportation and female safety in India. We chose women between the ages of 21-35 years old as our target user group because of the high rate of crime perpetrated against this group and the dire need to address gender equity in an emerging economy that is witnessing a rise of its young women entering the workforce. Our study further investigates the environmental and societal factors that make women feel unsafe and the familial management of women's safety that is unique to collectivist societies like India. In response to our research findings, our team created Navi.io, a product that not only makes women feel safer, but also empowers them to make better informed travel decisions.

2 DESIGN RESEARCH

2.1 Diary Study Methods

Our team wanted to understand (1) the day-to-day experiences that influence women in India's perception of safety and security, (2) environmental factors that contribute to women feeling unsafe, and (3) their responses to situations that make them feel unsafe. To answer these questions, we chose to run a diary study for several reasons. First, our team is based in the United States and has a limited understanding of what day-to-day life is like for women in India and so we must collect rich and high volumes of qualitative data. Second, the perception of safety is an admittedly ambiguous, ephemeral, and highly contextual phenomenon of measurement. And so, a 7-day diary study allows our team to collect rich and in-context data.

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Recognizing the diverse range of female experiences in a country as large as India, we scoped our study and relied on purposive sampling and screened participants to include 6 Indian women living in large cities in Southern India between the ages of 20–35, who are unmarried, college-educated, and middle-class.

Before the study, our team met with each of the participants individually to review study expectations. For seven days, each participant was asked to keep a diary of their experiences related to safety and traveling within India, which include writing about any safety issues or challenges that they faced. In addition to textual diary entries, participants were encouraged to document their experiences and environments with video and or pictures. To standardize high-quality data collection, we provided each participant with a template for reporting their diary entries and instructions for completing entries in a comprehensive way. The template included several writing prompts and sample diary entries for participants to reference. Some prompts include: (1) Please describe the location and environment in detail and were there any factors about the environment that made you feel particularly unsafe? (2) What were you trying to accomplish at this time? (3) Who was involved? (4) Can you describe in detail what happened? (5) How did you deal with the situation? (6) Please include pictures, video/audio recordings, or illustrations that help describe the experience. To ensure high diary entry completion rates, our team sent 2 daily messages (1 at the beginning of the day, and 1 at the end of the day) to each participant reminding them to complete their diary entries.

2.2 Semi-Structured Interview Methods

Our team conducted remote 1-hour semi-structured interviews with the same 6 diary study participants and scoped our interview questions along two emerging themes from the diary study: (1) transportation challenges and (2) female responses and familial management of female safety. During the semi-structured interviews, we asked participants to elaborate on individual diary entries of interest as well as asked questions like: (1) Can you describe the last time you felt unsafe while riding public transportation, and what was your reaction? (2) Can you describe the relationship you have with your family when it comes to your safety? (3) Can you recall the last time a family member got involved in your safety?

2.3 Thematic Coding of Interviews and Diary Entries

Interviews and Diary Entries were synthesized and thematically coded using Dovetail software. Interviews were recorded and transcribed using a transcription service. A codebook of relevant themes was created based on (1) inductive themes gathered from post-interview team debriefs (2) original research objectives, and (3) deductive themes categorized and found in initial literature review. The themes were then narrowed by team consensus to ensure that themes were comprehensive and relevant to our study. Finally, members of the team thematically coded the transcripts using the finalized codebook.

2.4 Diary Study and Semi-Structured Interview Findings

From our diary study and subsequent interviews with participants, we identified 3 high-level themes related to (1) transportation challenges, (2) female and familial responses to safety concerns and prevention methods, and (3) factors of the physical environment that make women feel unsafe.

2.4.1 Transportation Challenges. In the diary studies, all participants cited frequent challenges with unreliable public transit systems that cancel routes with no means of notifying passengers, causing women to wait hours, often at night, for buses and trains that will never arrive. One participant even reported that their bus canceled their route mid-route and deposited passengers in an unfamiliar location. Unreliable public transit is especially problematic during national holidays with high transit use, exacerbating traffic and transit delays.

The interviews also revealed that all the participants almost exclusively traveled within “women’s only” compartments on public transit that men are not allowed to enter. However, these compartments are often overcrowded since there are fewer compartments just for women.

For many of these women, public transit is their only affordable means of transportation. Some cities have government-owned apps that notify passengers of train delays, but many cities do not have them, and rarer still are tools that notify passengers of bus delays.

2.4.2 Female and Familial Responses to Female Safety. Most respondents reported instances where men have flashed and or inappropriately touched the participants on public transit or in the streets. During these incidents, two women reported feeling helpless, freezing up, and unsure how to respond.

Many participants report regularly rerouting their routes if they see that there are entering areas with high male traffic. 3 women report avoiding using public transportation alone as much as possible in favor of traveling with others and or relying on private transit options like having family members drive or driving themselves to their final destination.

Two participants have previously reported instances of sexual harassment to the police and school authorities, but they were largely ignored or unresolved. Another participant never reported incidents to authorities out of fear of retaliation or victim blaming, “One of my classmates reported that she had been assaulted by a classmate at the school, but they didn’t do anything about it and everyone victim-shamed her.”

For many of the participants, their families played a large role in ensuring the safety of the females in the family. Their parents set curfews before dark and frequently call to receive status updates about their daughters’ whereabouts in transit. One participant’s father went so far as to arrange for family friends living in neighboring cities to personally chauffeur the participant whenever she traveled to their city, out of concern for her safety when traveling alone in an unknown city. Due to family concern for female safety, some participants report rarely leaving their homes unless they are accompanied by a family member or close friend. Those who did travel alone, often shared their location via Whatsapp, with family members, while in transit.

While many respondents report feeling thankful that their parents are so involved in their safety, interviews with other respondents suggest that family involvement can sometimes feel restrictive, as highlighted in the following quote from one participant, “If I told my parents what happened, they would just be scared and then wouldn’t send me out, and I just lose my freedom because of it.” Some respondents report withholding information about their whereabouts to their parents, and instead “just tell them I’m doing something else that I enjoy,” as expressed by one respondent.

2.4.3 Factors of the physical environment that make women feel unsafe. Participants reported that many public areas are inaccessible and unsafe for women like smoking areas and liquor stores. These male-dominated spaces are hard to avoid, and respondents often feel unsafe because groups of men are staring at them when they pass. Furthermore, all participants reported traveling often through poorly lit and dark streets. These feelings of unsafety due to isolation in public spaces are further exacerbated at night when shops and other public spaces are closed.

2.5 Competitor Analysis Methods

To better understand the competitive landscape, we conducted a competitor analysis on five competitor products based in India: (1) Raksha (2) My Safety Pin (3) Himmat (4) bSafe (5) Shake2Safety. These apps were chosen because (1) their core users match our target audience who are women living in major Indian cities and (2) they had the largest active user base and the largest number of app downloads for mobile apps related to female safety and travel. Some applications were privately-owned, and others were state-owned. We audited each product’s core capabilities and reviewed the highest and lowest user reviews to identify must-have features and market gaps. Our primary variables of interest and comparison included (1) concern for user privacy and security (2) efficient interactions and ease of accessing help (3) social features that enabled community members to get involved in female safety.

2.6 Competitor Analysis Findings

The competitive analysis revealed that many competitors had limited capabilities beyond creating static textual content providing general advice on how to travel safely. Furthermore, the content and features were often not personalized to the user’s immediate location, limiting their efficacy. Although some competitors enabled users to share their location, it was unclear whether the competitors made provisions that allowed users to control and modify who has access to their location, raising data privacy concerns. We found that many competitor interventions required users to complete lengthy forms to file a report and many of these reports were sent to governmental institutions. It is unclear how users would benefit from filing reports through our competitors’ applications, given the pervasive cultural of victim blaming and institutional indifference towards female safety concerns. Even if individual reports did initiate institutional recourse, its impact is limited to helping the individual who filed the report, and others are unlikely to benefit, thereby raising a scalability concern. Furthermore, the unintuitive reporting process required too many steps for a user to complete, which would be inaccessible in a true emergency. These findings suggest a need to design a solution that both (1) reduces the friction

and time spent for individuals to file safety reports and (2) democratizes the process of reviewing reports so that a broader community of women can benefit.

3 DESIGN PROCESS

3.1 Product Requirements

Based on our population, we derived several table-stake requirements for our product: (1) Enable safety features for users while traveling (2) Quick and easy to use (3) Does not impact the safety of users (4) Low cost and accessible (5) Easily fit into user’s lifestyle.

3.2 Capabilities and Design Decisions

After collecting and synthesizing data from our research, we moved onto our ideation phase with the three recurring themes in mind: (1) transportation challenges, (2) familial management of female safety, and (3) environmental factors that make women feel unsafe. These themes informed our app’s three key design decisions: (1) a navigation system with safe route features, (2) easy location sharing with friends and family, and (3) reporting and sharing crowdsourced data.

3.2.1 Navigation System. Our research findings gathered from primary research with users validated our secondary research, which found that women experienced the most uncomfortable safety concerns in transit and using public transportation, especially when transit delays or cancellations occur. To ensure transportation is more reliable and transparent, we decided a core feature would be the use of crowdsourced data from other passengers to keep users informed with the most accurate live reporting.

While using Navi.io’s navigation system, users are updated when their bus or train is delayed or canceled using crowdsourced live updates from other users who are bus stations ahead of them [Figure 1]. This will provide users with a sense of safety and allow them to wait at safer locations before their rides arrive. Using historical and live data, the app also provides users with route recommendations that maximize safety without compromising on efficiency to arrive at destination. Users will also have access to view in-context reports made on the map and choose an alternative walking path that bypasses potentially dangerous situations.

3.2.2 Share My Location. After analyzing the discussions and family decisions related to safety that females have with their families, we found that families and friends play a large role in a user’s preventative safety measures when traveling, through the sharing of locations and travel details. We found an opportunity to maximize this existing positive behavior by simplifying the sharing of location and route with family and friends within Navi.io [Figure 2]. Users can share their location with multiple contacts at 3 different intervals to preserve user privacy: an hour, end of the day, or indefinitely. While in transit, users can also share their route with others to track their location. If the user takes longer than expected to reach their destination, their contacts are notified of the delay and are encouraged to check in on the user. This feature eases the effort of location sharing and allows both parties to have peace of mind.

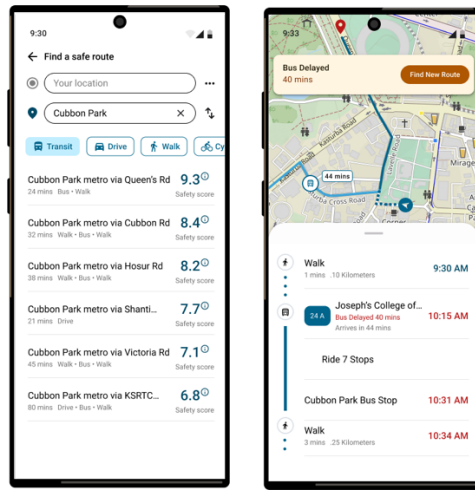


Figure 1: Navigation screens in Navi.io. Users can view and choose from routes to take with different safety scores (left). While en route, users are notified if there are any public transit delays or changes (right).

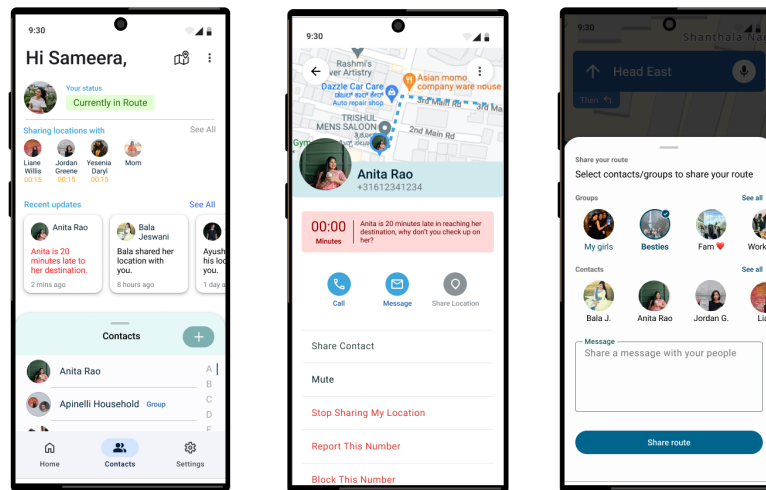


Figure 2: Navi.io's share my location feature. Users can easily choose who they would like to share their route with individually or in groups (left). Users can share their location with contacts and view the progress of their contact's journey (right).

3.2.3 Building a Community Through Shared Experiences. Our research informed us of common environmental and social factors that have made women feel less safe such as dim street lighting, male-centric areas, inappropriate touching, and delayed public transportation. This led us to the idea of embedding opportunities for reporting and viewing reports of incidents for users to feel more informed and if possible, prevent a similar occurrence from happening [Figure 3].

Through anonymized frictionless reporting, users can make a report within seconds to share their incident with others on Navi.io. This task was designed to be intuitive and seamless for the user to

want to repeat the task through a 3-tap survey. Users are motivated to continue reporting incidents when they are notified that many other users have seen and therefore benefit from their reports. This feature also formalizes a sense of community by showing that women are not alone in the challenges they face. It gives them a platform of anonymity that shields them from victim blaming while empowering them to voice their experiences.

3.3 Iterative User Testing

We recruited 5 female participants from Bangalore and Hyderabad to test Navi.io through moderated in-person and remote usability

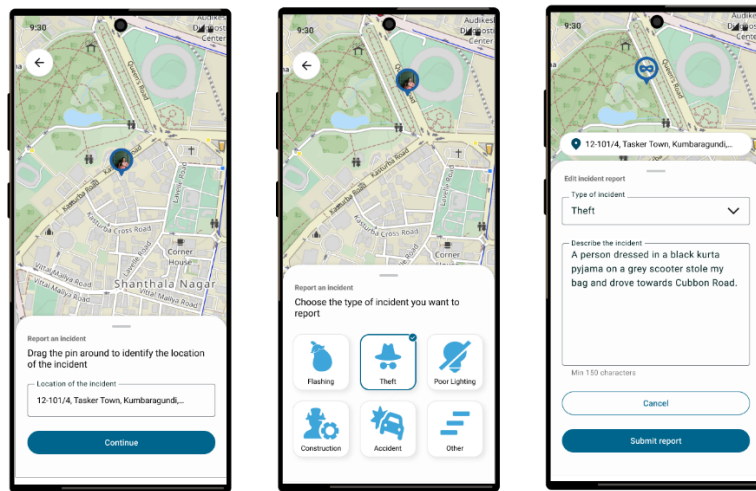


Figure 3: Navio's report incident feature. These are the three-step process to completing a report to share with the Navi.io Community.

tests of our app's core features. Per Nielsen Norman Group, 5 usability tests maximize ROI and capture around 85% of a product's usability concerns, and for our study, our team reached a saturation of redundant findings once we reached the sample size of 5 [4]. Participants were recruited through convenience sampling, between the ages of 21 and 26, with income ranging from 0 - 9,00,000 Rupees per annum. Two participants are students, two work in the software industry and one is a doctor. Participants are frequent users of either public transport or shared private transport.

Participants were given 7 tasks to complete without guidance paired with follow up questions and a final survey for assessment research. Overall, Users found the application easy to use with a system usability scale (SUS) score of 89 out of 100. With each screen assigned a criticality score, we were able to cross analyze the severity and frequency of errors to prioritize what we need to redesign.

One of the critical reoccurring usability issues was challenges with inputting location information for navigation. Some users had a mental model of moving the pin on the map while others were used to typing in the information. We adapted our final prototype to include both methods of inputting information and addressed additional minor usability concerns in the final iteration of Navi.io.

4 DISCUSSION

Recognizing the diverse range of female experiences in a country as large as India, we intentionally scoped our study to understand the experiences of a specific group of female users. Future work should include studying the needs of females who represent more age ranges, as well as socioeconomic and regional backgrounds. Furthermore, this product could prove valuable for other vulnerable populations within India like members of the LGBTQIA+ community, and so future work should explore the needs of members from this diverse community.

Furthermore, India has one of the largest gender gaps in female access to mobile devices and so more research should be done to examine alternative low-tech and non-tech solutions to ensuring female public safety [6]. This is especially true in rural communities in India and communities with lower education, lower caste, and poorer families.

Even among Indian women who do have access to mobile devices and can therefore use Navi.io, only seventy-one percent of Women in India are digitally literate. Although Navi.io relies on the use of iconography to convey many of the app's functionality, many areas within the app are still text heavy. More research should be done to identify opportunities to minimize the reliance on text content and to validate whether or not the iconography used in the app is universally understood by women in India.

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